

Date : 17.09.24	Roll. No:.....
ECECC06/EIECC06	
III SEMESTER-B.Tech.(ECE& EIOT)	MID-SEMESTER EXAM, SEPT., 2024
Course Code:ECECC303/EIECC303	Course Title:Prob. Th. and Random Process
Time: 1:30 Hours	Max. Marks: 20
Note: -For Re-registration ie ECECC06/EIECC06, marks shall be scaled up to 25 marks. Attempt all questions in the given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.	

S. No.	Questions	Marks	CO
1. (a)	Let A, B, and C be three events in the sample space S. If it is given that $A \cup B \cup C = S$, $P[A] = 1/2$, $P[B] = 2/3$, $P[A \cup B] = 5/6$, and $P[C \cap (A \cup B)] = 5/12$. Find a) $P(A \cap B)$. b) Do A, B, and C form a partition of S? c) Using $P(C - (A \cup B))$, find $P(C)$.	2	CO1
(b)	If we roll two dice and observe two numbers X and Y. Find a. $P(X=2, Y=6)$. b. $P(X>3 Y=2)$. c. If $Z=X+Y$, find the PMF of Z, $P_Z(z)$. d. Find $P(X=4 Z=8)$.	2	CO1
2. (a)	The PDF of a continuous r.v., X, is given by $f_X(x) = \begin{cases} x, & 0 < x \leq 1 \\ 2-x, & 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$ Find the corresponding CDF and sketch $f_X(x)$ and $F_X(x)$.	2	CO1
(b)	Let X be a Poisson r.v. with parameter λ . Derive the mean and variance of this random variable.	2	CO1
3. (a)	Let $X \sim N(0, \sigma^2)$. Determine $E[X X>0]$ and $\text{Var}[X X>0]$?	2	CO1
(b)	What does Moment Generating Function (MGF) signify? How can we obtain moments of a random variable using its MGF?	2	CO1
4. (a)	Consider the binary communication channel shown below Let (X, Y) be a bivariate r.v., where X is the input to the channel and Y is the output of the channel. Let $P(X=0) = 0.5$, $P(Y=1 X=0) = 0.1$, and $P(Y=0 X=1) = 0.2$. Find 1. Joint PMF of (X, Y). 2. Are X and Y independent?	2	CO2

(b)	Let X and Y be jointly continuous random variables with joint PDF $f_{XY}(x, y) = \begin{cases} 6e^{-(2x+3y)}, & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$ Find 1. $E[Y X>2]$. 2. $P[X>Y]$.	2	CO2
5.(a)	The joint PDF of random variables X and Y is given by $f_{XY}(x, y) = \begin{cases} k & 0 < y \leq x < 1 \\ 0 & \text{otherwise} \end{cases}$ Determine 1. the value of k. 2. Are X and Y independent? 3. Are X and Y uncorrelated?	2	CO2
(b)	Suppose the joint PMF of a bivariate r.v. (X, Y) is given by $P_{X,Y}(x_i, y_j) = \begin{cases} \frac{1}{3} & (0,1), (1,0), (2,1) \\ 0 & \text{otherwise} \end{cases}$ 1. Compute $E[X]$, $E[Y]$. 2. Covariance of X and Y.	2	CO2